ENERGYWATCH — SMART HOME ENERGY MONITORING & BILL FORECASTING DASHBOARD
PROJECT REPORT, PRESENTATION SCRIPT & TECHNICAL DOCUMENTATION

1. PROJECT OVERVIEW

Project Name: EnergyWatch
Target Domain: Smart Home Energy Management, IoT Energy Telemetry & Green Computing
Frameworks & Tech: Flutter (Material 3), Vanilla HTML5, CSS3 Custom Properties, JavaScript
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Live Calibrated Metrics:

- Total Monthly Baseline Consumption: 184.5 kWh
- Today's Energy Consumption: 6.72 kWh (5.4% reduction vs yesterday's 7.10 kWh)
- Estimated Monthly Electricity Bill: ₹1,576.00 (at ₹8.00/unit base tariff)
- Potential Monthly Savings: ₹286.00 / month
- Conserved Energy Potential: ~35.8 kWh / month
- Carbon Offset: ~29.3 kg CO2 / month
- Live Active Power Telemetry: 1.84 kW (realistic residential drift: 1.70 - 2.15 kW)

2. PROBLEM STATEMENT & SOLUTION

Core Problem:
Traditional utility bills arrive 3 to 4 weeks after electricity has been consumed.
Consumers have zero real-time visibility into:

1. Instantaneous power draw across household appliances.
2. Disaggregation — which specific appliance (AC, Geyser, Refrigerator) drives peak costs.
3. Realistic end-of-month bill forecasts before it is too late to make adjustments.
4. Specific, measurable habits to lower both expenses and carbon emissions.

EnergyWatch Solution:

1. Live Active Demand Monitoring: Visualizing real-time active load in kW.
2. Consumption History & Peak Hour Highlighting: 24-Hour load profile showing residential evening peak hours (6:00 PM – 10:00 PM) in amber warnings.
3. Appliance-Level Tracking: 10 appliances tracked with power ratings, daily runtimes, monthly kWh, efficiency ratings, and an interactive runtime adjustment simulator.
4. Dual Tariff Bill Estimator: Supports both Flat Single Rate and Progressive Tiered Slab Billing with line-by-line itemized computation.
5. Actionable Savings Engine: 6 prioritized habits with step-by-step guidance and dynamic cost/CO2 reduction tracking.

3. HARDWARE-READY DECOUPLED ARCHITECTURE

Architecture Pipeline:
[Physical IoT Sensors] --> [Microcontroller] --> [Network Transport] --> [EnergyDataSource] --> [EnergyRepository] --> [UI Layer]
(SCT-013 CT Clamp / PZEM-004T AC Module) (ESP32 / ESP8266) (MQTT Broker / REST) (Abstract Contract) (Single Source of Truth) (Flutter / Web)

Decoupling Principle:
The software architecture implements the Repository Pattern with an abstract EnergyDataSource.
Because physical sensors were simulated for this presentation prototype, future hardware integration requires only writing a single ApiEnergyDataSource class. The UI, state management,
charts, and calculations require ZERO code modifications.

(-2.8%).
4. LED Lighting Upgrades: Replace CFLs and use natural daylight. Saves ₹38/mo (-2.5%).
5. Refrigerator Coil Maintenance: Clean rear coils & ensure 10cm gap. Saves ₹24/mo (-1.6%).
6. Peak Load Shifting: Run laundry/pumps during off-peak morning hours. Saves ₹15/mo (-1.0%).
TOTAL MONTHLY POTENTIAL SAVINGS: ₹286 / month (19.4% overall reduction)

7. MATHEMATICAL FORMULAS
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1. Appliance Daily Energy:
Energy (kWh) = Power (kW) × Runtime (hours)

2. Appliance Percentage Share:
Share (%) = (Appliance Monthly kWh / Total Household Monthly kWh) × 100

3. Flat Rate Electricity Bill:
Energy Charge = Consumption (kWh) × Rate (₹/kWh)
Tax Charge = Energy Charge × (Tax % / 100)
Total Bill = Energy Charge + Fixed Meter Charge + Tax Charge

4. Progressive Tiered Slab Bill (Utility Standard):
For each Tier_i with bounds [Min_i, Max_i] and Rate_i:
Units_in_Tier = min(Remaining_Units, Max_i - Min_i)
Tier_Cost = Units_in_Tier × Rate_i
Total_Energy_Charge = sum(Tier_Cost)
Total Bill = Total_Energy_Charge + Fixed Charge + Tax Charge

5. Potential Monthly Cost Savings:
   Energy_Saved (kWh) = Monthly_Consumption × (Reduction_% / 100)
   Cost_Savings (₹) = Energy_Saved × Rate_per_kWh

6. Carbon Dioxide Emission Offset:
   CO2_Offset (kg) = Conserved_kWh × 0.82 kg CO2/kWh (Grid average factor)

8. SOURCE CODE FILE DIRECTORY MAP

Flutter Architecture:
- lib/main.dart --> App entry point & provider injection
- lib/core/constants/ --> AppConstants, slab tiers definitions
- lib/core/theme/ --> AppColors, AppTheme (Material 3 Light/Dark)
- lib/core/utils/calculations.dart --> Pure mathematical formulas & billing models
- lib/core/utils/formatters.dart --> Currency, energy, date/time formatters
- lib/models/ --> Appliance, EnergyReading, BillingRate, SavingsTip
- lib/providers/energy_provider.dart--> ChangeNotifier state & live simulation timer
- lib/repositories/ --> EnergyRepository (Repository pattern)
- lib/data/mock_energy_data.dart --> 30-day calibrated telemetry dataset
- lib/screens/ --> Dashboard, Consumption, Appliances, Details, Bill, Savings, Settings
- lib/widgets/ --> StatCard, ConsumptionChart, ApplianceCard, UsageProgress, etc.

Plain Web & Figma Export Architecture:
- build/web/index.html --> Standalone semantic HTML5 web application
- build/web/style.css --> CSS custom variables, dark mode, card designs
- build/web/app.js --> Client-side state engine, SVG charts, simulator
- build/web/energywatch_figma.html --> Inlined 5-screen single file export for Figma
- screenshots/ --> 7 clean high-resolution presentation screenshots